

Text Summarization



Florina Piroi

Content

- Introduction
- Processing & baseline algorithms
- Evaluation
- Very active research area

Introduction

Motivation for Automatic Text Summarization

- Lots and lots of documents, mass of information, unstructured text
 - “too much information kills information”
- No standards in how to process them

Summaries written by:

- document author
- Professional summarizer
- Third party

Writing a summary is difficult – cognitive process!

select, comprehend (and reformulate), stay coherent

consistency (same person, different summaries at different times)

Challenges in automating it

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Automatic Text Summarization

- Condense the text
- Maintain relevant information
- Limited understanding of the text

Why (Automatically) Summarize?

- Reduce reading time
- Selection process is easier
- Effective indexing (by librarians)
- Automatic summarization is less biased
- Useful in Question-Answering systems
- Enable service providers to process more docs

What is a Text Summarization?

- 1979 (**ANSI**) : [An abstract] is an abbreviated, accurate representation of the contents of a document, preferably prepared by its author(s) for publication with it. Such abstracts are useful in access publications and machine-readable databases.
- 1980 (van Dijk): The primary function of abstracts is to indicate and predict the structure and content of the text.
- 1973 (Cleveland): An abstract summarizes the essential contents of a particular knowledge record, and it is a true surrogate of the document.

Produced by people!

What is a *Automatic* Text Summarization?

- OED (Oxford English Dictionary): The creation of a shortened version of a text by a computer program. The product of this procedure still contains the most important points of the original text.
 - 2001 (Spärk-Jones, Sakai): A summary is a **reductive transformation** of a source text into a summary text by extraction or generation.
- > compression, loss of information
- Which information to include in a summary?
 - Important & representative content (how do you measure that?)

What is a *Automatic* Text Summarization?

- OED (Oxford English Dictionary): The creation of a shortened version of a text by a computer program. The product of this procedure still contains the most important points of the original text.
- 2005 (Hovy): An automatic summary is a text generated by a software, that is coherent and contains a significant amount of relevant information from the source text. Its **compression rate τ** is less than a third of the length of the original document.

$$\tau = \frac{|Summary|}{|Source|}$$

10%, 15% - 30%

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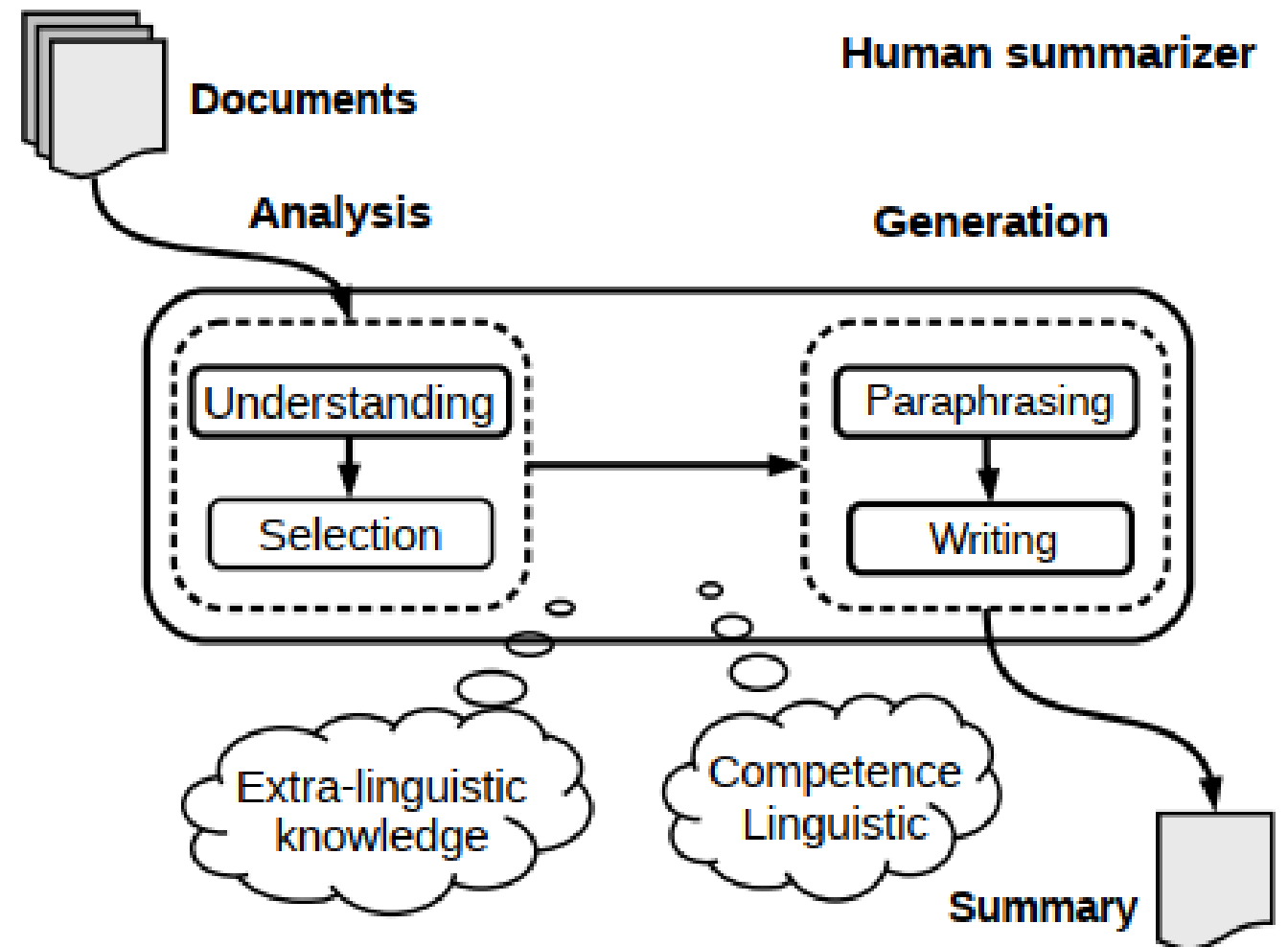
Generating a Summary?

- “Art of Abstracting” – Edward Cremmins (1996)
- Highly structured and coherent
- Introduction (1-2 general sentences, 2-3 specific)
- Aims (1 sentence)
- Methods (1-2 sentences)
- Results (3-4 sentences)
- Conclusions (1-2 sentences)

(Summarizer need not be an expert, 8-12 minutes, no deep understanding)

(Prerequisite: papers follow a certain structure, content quality)

- Should we replicate this?
- Probably not ...
- ... but exploitable properties!



Different aspects of Summaries

(High-level) Objectives:

- Direct objective (why):
 - Give overview of the document
 - Update the user
 - Eliminate language barriers
 - Information Retrieval
- Indirect objective (what/how):
 - Classify, index
 - Extract keywords
 - ...

Summaries ... by their function

- Indicative summary
 - Give info about the topics discussed
 - Resembles a table of contents
- Informative summary (much harder)
 - Reflect the content of the source text
 - Explaining the arguments
 - Shortened version of the document

Summaries ... by number of documents

- Single-document
- Multi-document
 - Not always heterogeneous
 - Usually on the same topic

Summaries ... by genre

- News summary (news articles)
- Specialised (on a specific domain – law, science, technology)
- Literary (narrative summaries, etc.)
- Encyclopaedic (summary of encyclopaedic docs)
- Social networks (of blogs, very short docs, etc.)

Summaries ... by type

- Extract
Fragments from the original document
- Abstract
Reformulating/paraphrasing
- Sentence compression
Same number of sentences as in the original doc, but shorter (simpler?)

Summaries ... by type of summarizer

- Author summary
Written by the author
- Expert summary
Person other than the author, not an expert summarizer
- Professional summary
Person does not specify in the field, but knows how to summarize

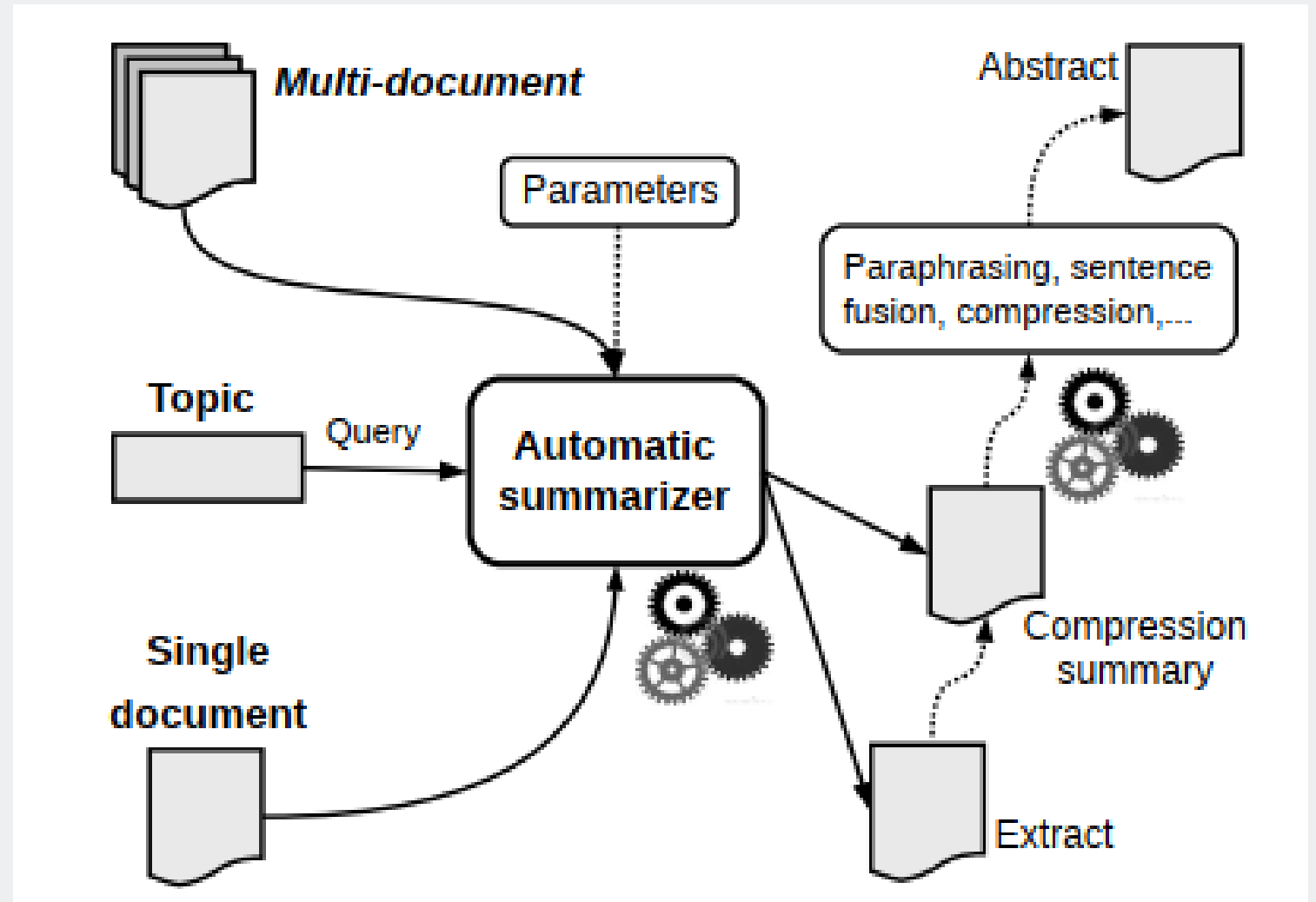
Summaries ... by context

- Generic summary
Ignores information need
- Query guided summary
Guided by some user information need/query
- Update summary
User familiar with topic, show only new, important information

Summaries ... by target audience

- Without a profile
Independent of the user characteristics – only based on document
- User profile
Targeted at users interested in particular domains (phys, chem, etc)

Generating automatic text summaries



Applications

- Better IR and IE performance
- Q&A
- Newswire generation
- RSS feed summarization
- E-mail and e-mail thread summarization
- Meeting summaries
- Automatic generation and extraction of titles
- Opinion summarization
- Domain-specific summarization (medicine, law, etc).
- ...

Processing & Baseline Summarization Algorithms

Extraction vs. Abstraction vs. Compression

- Summarization by
 - Extraction
 - Abstraction
 - Sentence compression

Pre-processing: Before the Summary

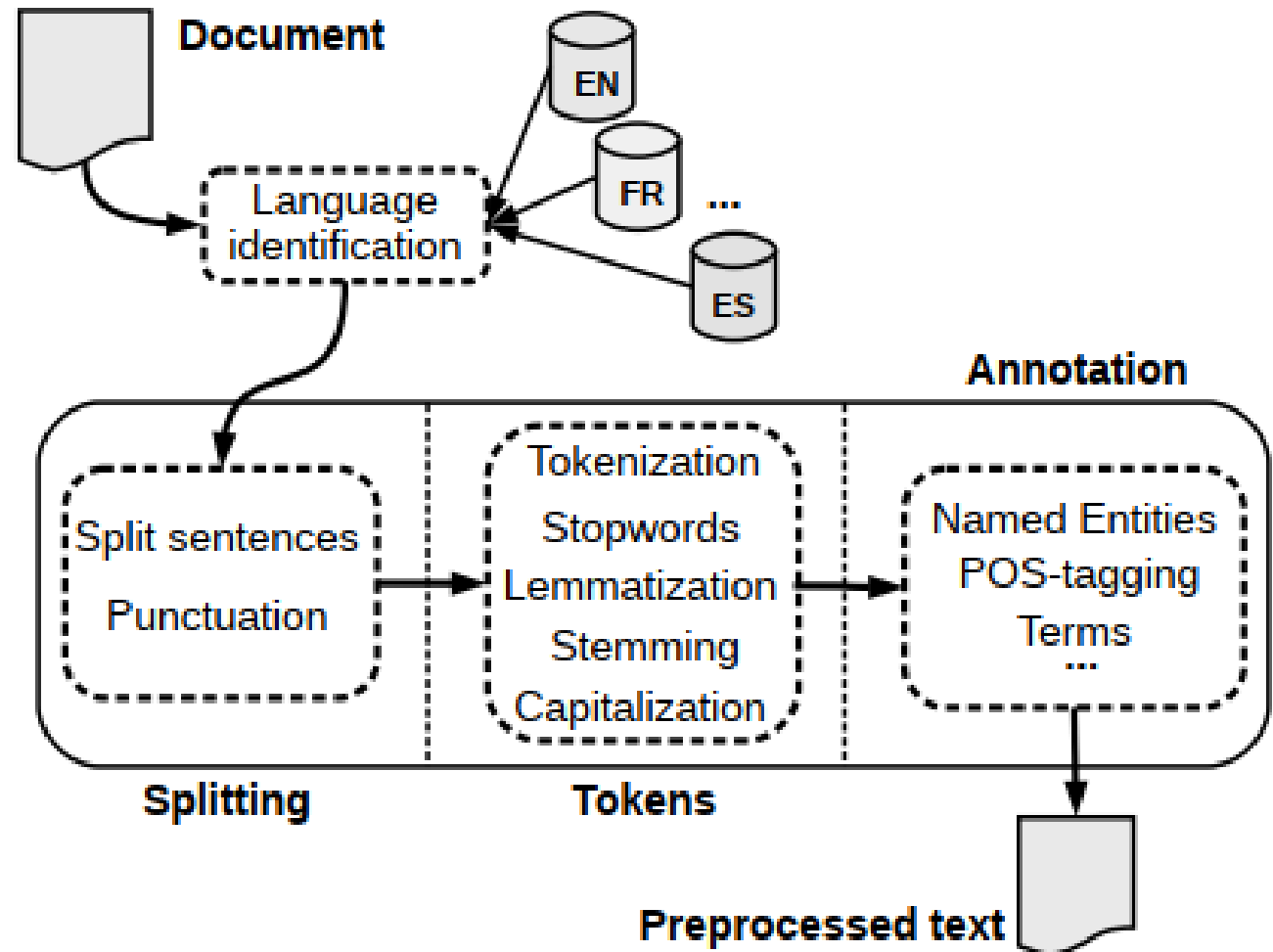
- Objectives
 - Word normalization
 - Lexicon reduction
- Essential to providing clean, adequate representations of the source documents.

Pre-processing: Before the Summary

- (Standard) stages:
 - Split into segments / paragraphs / sentences / etc.
 - Split segments into words / tokens
 - Normalization: lemmatization, stemming, etc.
 - Filtering (stop word removal)
- Optional:
 - Annotations (e.g. POS)
 - Named Entity recognition (NE)
 - Term extraction (keywords)
 - Weighting (tf-idf, embedding vectors, etc.)

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Extraction vs. Abstraction vs. Compression

- Summarization by
 - Extraction
 - Abstraction
 - Sentence compression (~33%)
- Summary: reduced representation, keeps essential content of a text
- Abstract: reformulating sentences (“good student”)
- Extract: picking out (extracting) pieces of the original text (“bad student”).
- Misuses leading to confusion & ambiguity: Zusammenfassung, Résumé

Extraction vs. Abstraction vs. Compression

- Select units of texts based on informativity
 - Assemble them in a useful way.
 - Aim: give an overview of the original text's content.
 - Length: determined by compression rate.
-
- (2000) 81.5% sentences borrowed verbatim (manually generated summaries)
 - (2004) 70% sentences in summaries appear in the original text

Extraction vs. Abstraction vs. Compression

- (Radev et al 2002) Algorithms are:
 - Surface-level
 - No linguistic processing
 - Word weighting
 - Used for docs with a fixed structure
 - Segment position
 - Words / Key sequences (“it is important to stress that ...” / “for example ...”)
 - Intermediate-level
 - Deep parsing techniques

Extraction vs. Abstraction vs. Compression

- (Radev et al 2002) Algorithms are:
 - Surface-level
 - Intermediate-level
 - Use linguistic information
 - Not as complex as deep parsing algorithms
 - Use lexical chains – the strong chains go into the summary.
 - Anaphoric references
 - Deep parsing techniques

Extraction vs. Abstraction vs. Compression

- (Radev et al 2002) Algorithms are:
 - Surface-level
 - Intermediate-level
 - Deep parsing techniques
 - Use in-depth linguistic techniques
 - Exploit discursive structure theory (rhetorical structure theory)
 - Divide text into discursive (rhetorical) units
 - Apply weights to these units
 - Highest weighed units are selected for the summary

Extraction vs. Abstraction vs. Compression

- Final objective in Summarization
- Text understanding
- Generate correct & coherent text
- Can combine IE, IR and Natural Language Generation techniques

Extraction vs. Abstraction vs. Compression

- **Sentence compression** idea:
 - Eliminate non-essential information in a sentence
 - Preserve correct grammar
 - All sentences kept
 - No solution to it, yet.
- Applications: generate titles, subtitles for media, display info on mobile devices
- Compression rate up to 33% according to some authors.

Extraction vs. Abstraction vs. Compression

- **Sentence compression** idea:
 - Eliminate non-essential information in a sentence
 - Preserve correct grammar
 - All sentences kept
 - No solution to it, yet.
- **Multi-sentence fusion** (multi-sentence compression):
 - more documents
 - Redundant sentences indicator for importance
 - Sentences from different docs fused and compressed
 - Form of paraphrasing

Limits of Extraction

- Cohesion
- Coherence

(hint on how to evaluate the quality of a summary)

Limits of Extraction

- Cohesion

Cohesion between the sentences of a text is related to the absence of anaphora and unresolved temporal references in the generated summary.

- 1) I like cherries

- 2) I am allergic to them.

- Resolving anaphora in text is difficult

Limits of Extraction

- Coherence

The coherence of the summary is the absence of contradictions and/or redundancy in the sequence of sentences in a document.

Source: The aircraft manufacturing industry is still in crisis in 2011. In 2010 Airbus faced many problems due to financial difficulties and was forced to halt production of its new long-haul carrier A380. Transatlantic, its Boeing rival, has finally started production of its B770. The new B770 has received a large number of pre-orders from Asian airline companies. This is good news for the manufacturer who may end up owning a large share of the aircraft industry.

Summary: In 2010 Airbus faced many problems due to financial difficulties and was forced to halt production of its new long-haul carrier A380. This is good news for the manufacturer who may end up owning a large share of the aircraft industry.

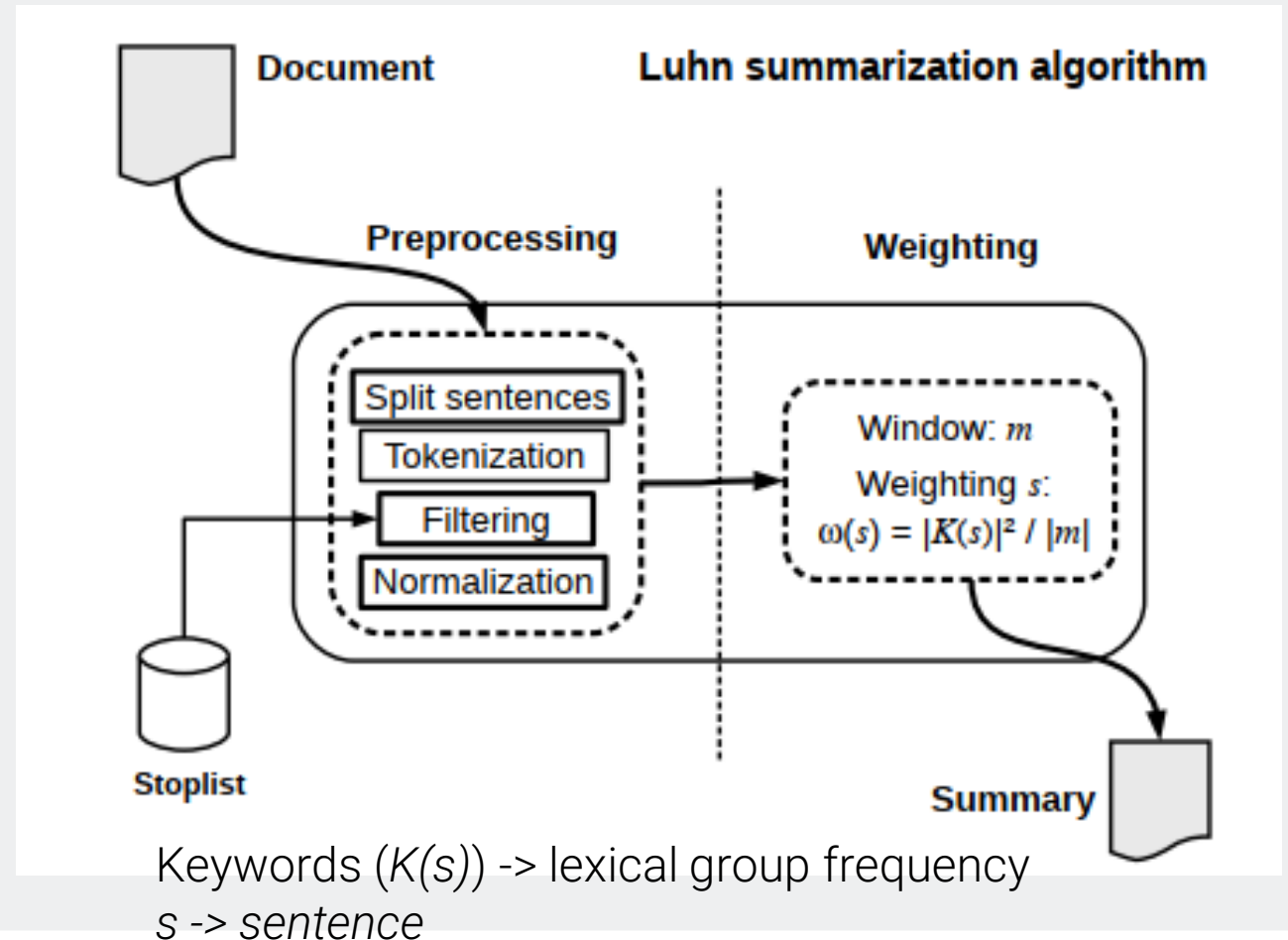
Algorithms

Algorithms: Luhn's Automatic Creation of Literature Abstracts

- Hans Peter Luhn (1896 – 1964)
 - Pioneer in automatic text summarization
 - 1958 (while at IBM) developed a simple algorithm to extract abstracts from scientific articles
 - Was worried about the information overload.
 - Recognized his work was “miles away” from text understanding & linguistics

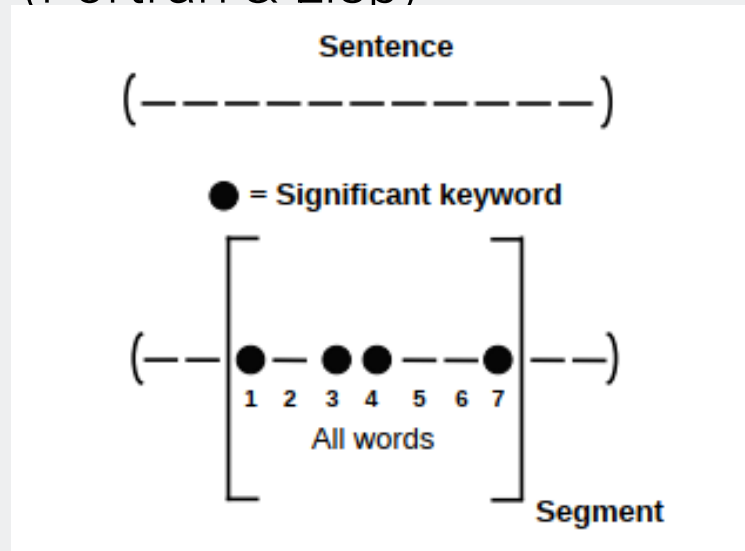
Algorithms: Luhn's Automatic Creation of Literature Abstracts

- Two phases:
 - Preprocess source document
 - Sentence weighing
 - (Fortran & Lisp)

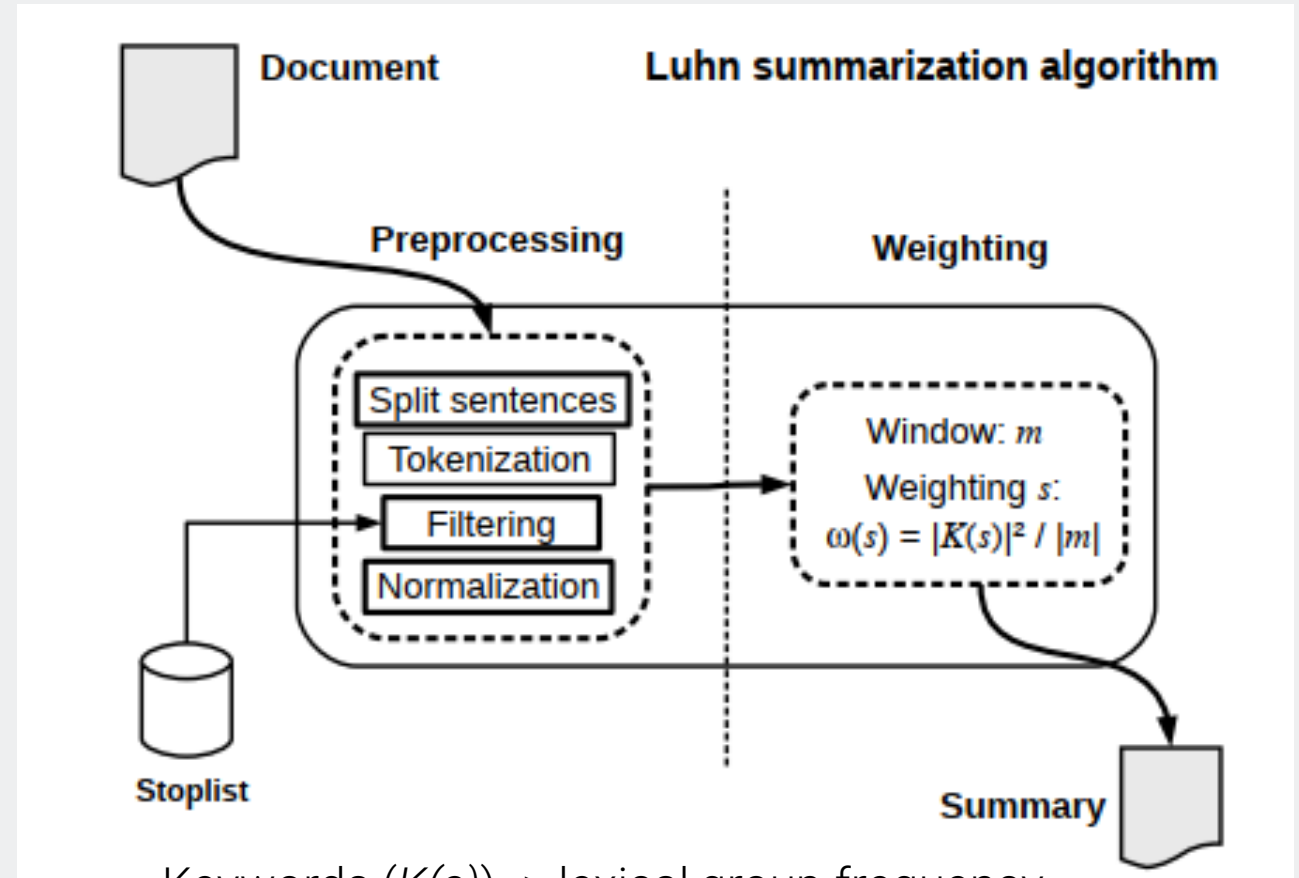


Algorithms: Luhn's Automatic Creation of Literature Abstracts

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- Sentence weight: 16 / 7



Keywords ($K(s)$) $\rightarrow$ lexical group frequency
 $s \rightarrow$ sentence

Algorithms: Graph-based Approaches

- Graph models to represent textual information:
 - Semantic disambiguation, POS tagging, IE, opinion analysis, etc.
 - Vertices (nodes) are semantic/cognitive units (words)
 - Edges are relations between units.
- Graphs -> *very sparse* binary matrix representation
- Effectively used in e-mail thread summarization, multisentence compression, tweet summarization, keyword detection, etc.

Algorithms: Graph-based Approaches

- PageRank
 - Computer representation for sparse matrices
 - Vertex centrality (from SNA)
- (central to IR – webpage ranking)
- Popularity of page i computed wrt. links starting/ending from/on i

$$p_R(i) = \frac{1-d}{N} + d \sum_{j=1}^N \frac{p_R(j)}{C(j)}$$

Algorithms: Graph-based Approaches

- LexRank & TextRank
 - Widely used
 - Based on the PageRank algorithm
 - Apply the graph to the concept of sentence weighting
 - Identify the most “prestigious” sentence in the graph!
- Document as graph
- Nodes are text units (sentences)
- Edges are similarity values between text units.
- “Central” sentences make it to the summary

Algorithms: Graph-based Approaches

- LexRank & TextRank
 - Document as graph
 - Nodes are text units (sentences)
 - Edges are similarity values between text units.
 - “Central” sentences make it to the summary
- Two algorithm stages:
 - Graph construction (sentences + similarity scores)
 - Link analysis (sentence “recommendation”)

Algorithms: Graph-based Approaches

- Graph construction:
 - Nodes: take whole sentences (simplest)
 - Edges: morphological similarity (~recommendation)
 - Cosine, word overlap, etc.
- $G = (V, E), E \subseteq V \times V, In() \text{ \& } Out()$
- Given $V_i, s_j \in In(s_i)$
- $\sum_{s_k \in Out(s_j)}$

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SNo	ID	Text
1	d1s1	Iraqi Vice President Taha Yassin Ramadan announced today, Sunday, that Iraq refuses to back down from its decision to stop cooperating with disarmament inspectors before its demands are met.
2	d2s1	Iraqi Vice president Taha Yassin Ramadan announced today, Thursday, that Iraq rejects cooperating with the United Nations except on the issue of lifting the blockade imposed upon it since the year 1990.
3	d2s2	Ramadan told reporters in Baghdad that "Iraq cannot deal positively with whoever represents the Security Council unless there was a clear stance on the issue of lifting the blockade off of it.
4	d2s3	Baghdad had decided late last October to completely cease cooperating with the inspectors of the United Nations Special Commission (UNSCOM), in charge of disarming Iraq's weapons, and whose work became very limited since the fifth of August, and announced it will not resume its cooperation with the Commission even if it were subjected to a military operation.
5	d3s1	The Russian Foreign Minister, Igor Ivanov, warned today, Wednesday against using force against Iraq, which will destroy, according to

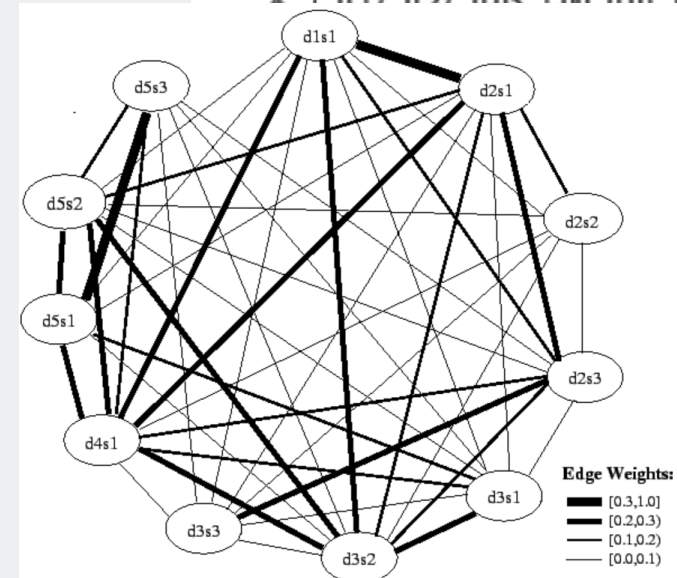
	1	2	3	4	5	6	7	8	9
1	1.00	0.45	0.02	0.17	0.03	0.22	0.03	0.28	0.04
2	0.45	1.00	0.16	0.27	0.03	0.19	0.03	0.21	0.04
3	0.02	0.16	1.00	0.03	0.00	0.01	0.03	0.04	0.04
4	0.17	0.27	0.03	1.00	0.01	0.16	0.28	0.17	0.04
5	0.03	0.03	0.00	0.01	1.00	0.29	0.05	0.15	0.21
6	0.22	0.19	0.01	0.16	0.29	1.00	0.05	0.29	0.04
7	0.03	0.03	0.03	0.28	0.05	0.05	1.00	0.06	0.04
8	0.28	0.21	0.04	0.17	0.15	0.29	0.06	1.00	0.21
9	0.04	0.04	0.04	0.04	0.21	0.04	0.04	0.21	1.00

Algorithms: Graph-based Approaches

- Graph construction:
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3	0.02	0.16	1.00	0.03	0.00	0.01	0.03	0.04	0.04
4	0.17	0.27	0.03	1.00	0.01	0.16	0.28	0.17	0.04
5	0.03	0.03	0.00	0.01	1.00	0.29	0.05	0.15	0.21
6	0.22	0.19	0.01	0.16	0.29	1.00	0.05	0.29	0.04
7	0.03	0.03	0.03	0.28	0.05	0.05	1.00	0.06	0.04
8	0.28	0.21	0.04	0.17	0.15	0.29	0.06	1.00	0.21
9	0.04	0.04	0.04	0.04	0.21	0.04	0.04	0.21	1.00



Algorithms: Graph-based Approaches

- Sentence weighting
 - $G = (V, E), E \subseteq V \times V$
 - $V_i, s_j \in In(s_i)$
 - $\sum_{s_k \in Out(s_j)}$
- LexRank:
 - (multidoc)

$$\omega(s_i) = \frac{d}{N} + (1 - d) \sum_{s_j \in In(s_i)} \frac{\text{sim}(s_i, s_j)}{\sum_{s_k \in Out(s_j)} \text{sim}(s_k, s_j)} \omega(s_j)$$

Algorithms: Graph-based Approaches

- Sentence weighting
 - $G = (V, E), E \subseteq V \times V$
 - $V_i, s_j \in In(s_i)$
 - $\sum_{s_k \in Out(s_j)}$
- TextRank:
 - (single doc)

$$\omega(s_i) = (1 - d) + d \sum_{s_j \in In(s_i)} \frac{w_{j,i}}{\sum_{s_k \in Out(s_j)} w_{jk}} \omega(s_j)$$

$$w_{j,i} = \frac{\sum_{w \in i,j} C_w^i + C_w^j}{\log |s_i| + \log |s_j|}$$

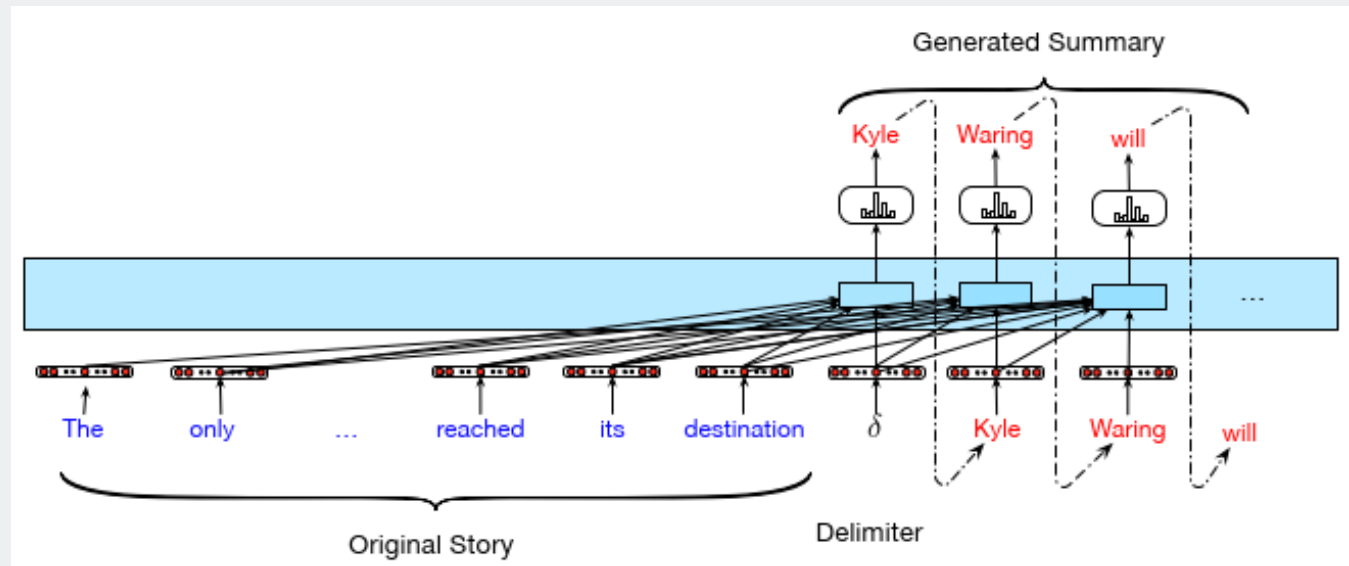
C occurrence of word w in sentence i
<http://lexrank.herokuapp.com/index.html>

Algorithms: Machine Learning Approaches

- Position of keywords -> weight of sentences -> selection into summary
- Use other parameters (features), too
- How to decide their contribution to the selection of a sentence?
- ML to estimate parameter values from their occurrences on a previously *tagged* corpus.
 - {article, summary}
 - Train a Bayes classifier to estimate sentence probability
 - Features: sentence length cut-off, fixed-phrases, position of sentence in paragraph, lexicon feature, upper case words,
 - (Kupiec et. al. 95 – small data set, 188 pairs)

Algorithms: Machine Learning Approaches

- Deep Learning methods
- Transformer-based autoregressive models
- Training data: {article, separator, summary}
 - $(a_1, \dots, a_m, \text{SEP}, s_1, \dots, s_n)$
 - Handled as long sequences



Algorithms: Machine Learning Approaches

- Deep Learning methods
- Transformer-based autoregression
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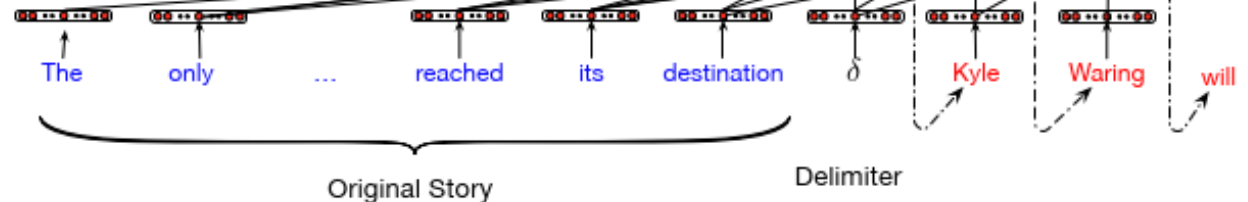
You

summarize this: The aircraft manufacturing industry is still in crisis in 2011. In 2010 Airbus faced many problems due to financial difficulties and was forced to halt production of its new long-haul carrier A380. Transatlantic, its Boeing rival, has finally started production of its B770. The new B770 has received a large number of pre-orders from Asian airline companies. This is good news for the manufacturer who may end up owning a large share of the aircraft industry.



ChatGPT

In 2011, Airbus faced financial issues, pausing A380 production, while Boeing's B770, receiving significant pre-orders from Asian airlines, is poised to dominate the aircraft industry.



Evaluation

Evaluation

- Input (text corpus) → Text Summarizer → Output (text summary)
- The little prince visited a planet inhabited by a drunkard. He talked to the drunkard and asked him why he drank. The response he received left him perplexed.
- The little prince visited a planet inhabited by a drunkard. After several bizarre answers from the drunkard, the little prince understood that the drunkard drank to forget that he was ashamed of drinking. The visit was very short and the little prince went away perplexed.

No perfect summary

Evaluation

- Intrinsic
 - measures the system's performance on its own (i.e. its output only)
 - Against human reference or gold standard reference
 - Baseline summaries:
 - Randomly extract n sentences
 - Use lead sentences, first n sentences
 - Use lead-end sentences, first n and the last m
 - Task specific baselines are also used.
- Extrinsic
 - how summaries are good enough to accomplish the purpose of some other specific task (IR, report generation, etc)

Semi-automatic Summary Evaluation

- Levels of granularity
 - Sentence
 - Word

Semi-automatic Summary Evaluation

- Sentence level evaluation:
 - Have a set of “perfect” summaries (*reference* summaries)
 - Precision & Recall for content overlap
- Precision: how many sentences match the ones in the reference summary?
- Recall: how many sentences were “forgotten” by a system
- F: harmonically combines P and R

$$P = \frac{|\text{Sum}_{\text{ref}} \cap \text{Sum}_{\text{can}}|}{|\text{Sum}_{\text{can}}|}$$

$$R = \frac{|\text{Sum}_{\text{ref}} \cap \text{Sum}_{\text{can}}|}{|\text{Sum}_{\text{ref}}|}$$

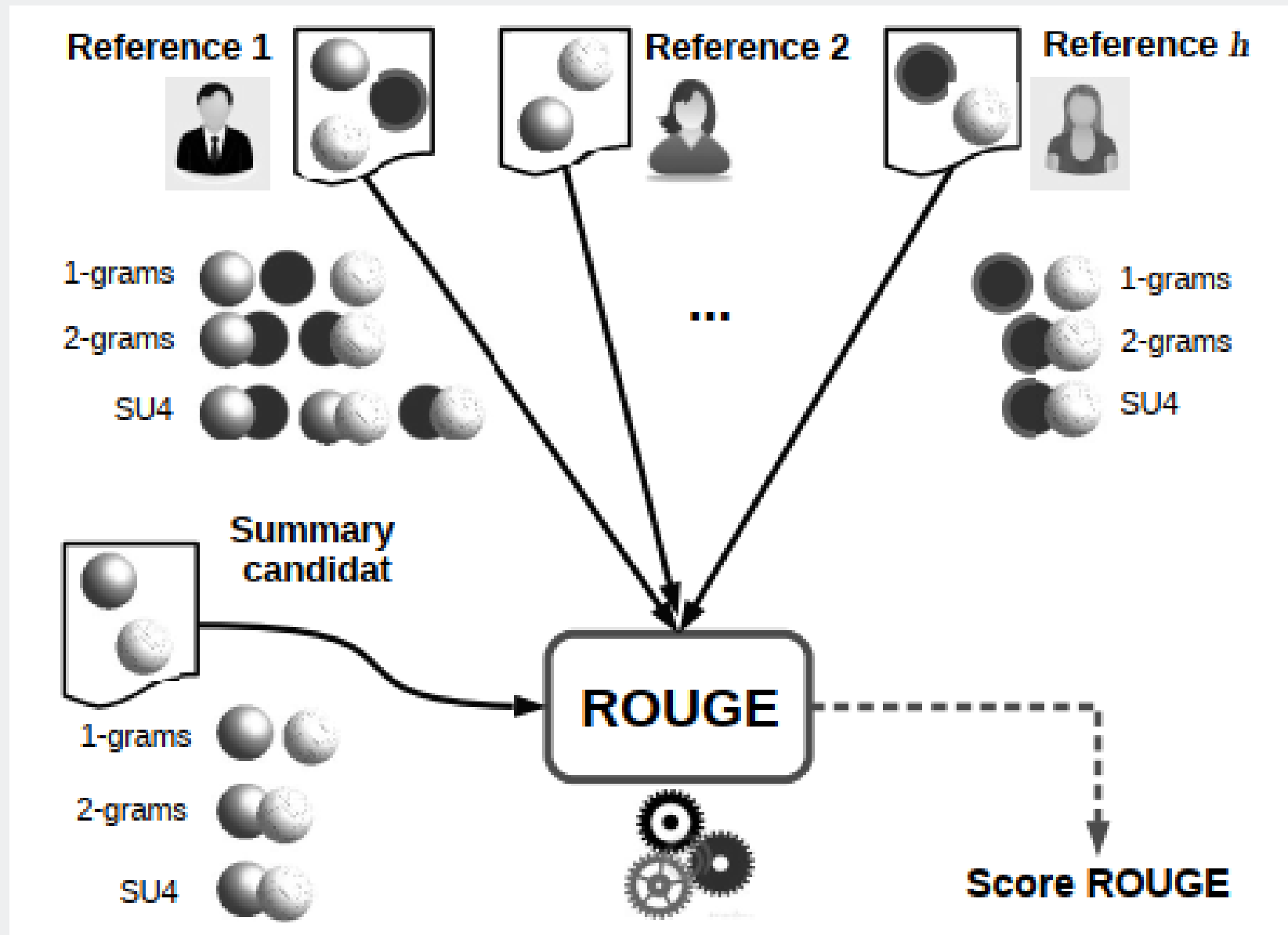
$$F = 2 \cdot \frac{P \cdot R}{P + R}$$

Semi-automatic Summary Evaluation

- Word level evaluation:
 - ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
 - PYRAMID
 - BASIC ELEMENTS (BE)
- Measure similarity to a reference summary.
- ROUGE (2004): look at difference in **word distribution** between the candidate and the reference summary.
- N-grams!

ROUGE

- Inspired by BLEU



ROUGE

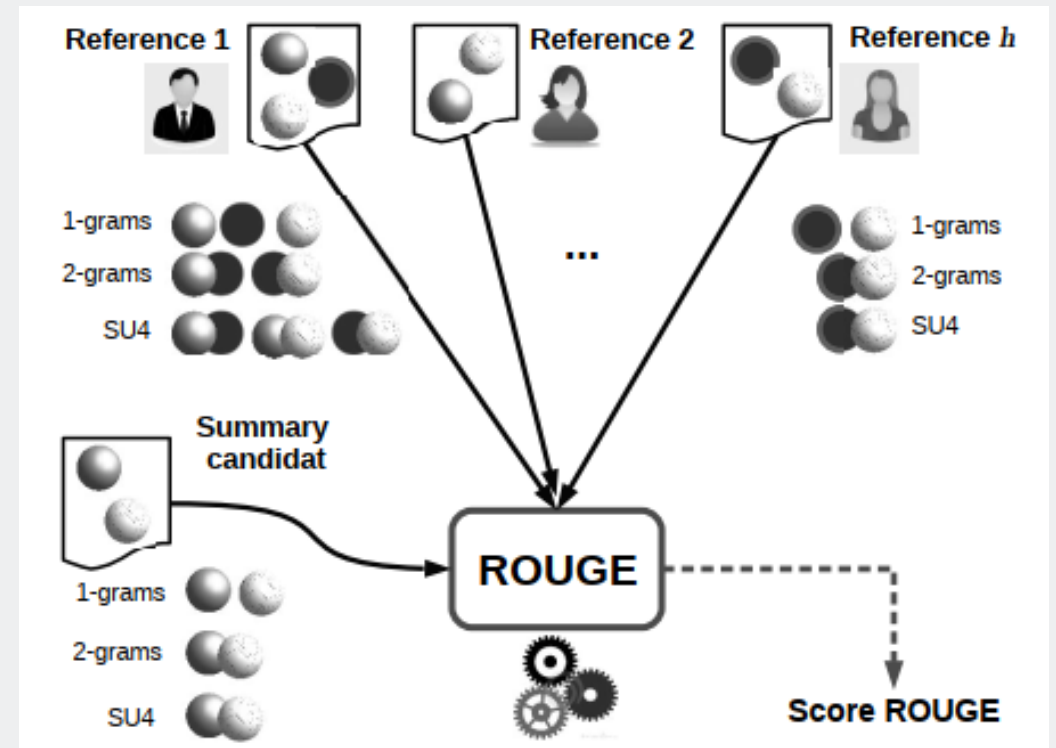
- Number of n-grams to be used in the Rouge-n metric:

Sentence	<i>Intellectuals solve problems; geniuses prevent them.</i>	Count
ROUGE-1	Intellectuals ▷ solve ▷ problems ▷ geniuses ▷ prevent ▷ them	6
ROUGE-2	Intellectuals solve ▷ solve problems ▷ problems geniuses ▷ geniuses prevent ▷ prevent them ▷	5
ROUGE-SU4	Intellectuals solve ▷ Intellectuals problems ▷ Intellectuals geniuses ▷ solve problems ▷ solve geniuses ▷ solve prevent ▷ problems geniuses ▷ problems prevent ▷ problems them ▷ geniuses prevent ▷ geniuses them ▷ prevent them	12

ROUGE

- Rouge-2, Rouge-SU4 mostly used
- Rouge-SU γ ~ ROUGE-2 using skip units (SU) of size $\leq \gamma$
- Rouge-SU4 – max window length is 4.

$$\text{ROUGE-}n = \frac{\sum_{n\text{-grams} \in \{\text{Sum}_{\text{can}} \cap \text{Sum}_{\text{ref}}\}}}{\sum_{n\text{-grams} \in \text{Sum}_{\text{ref}}}}$$



ROUGE-SU γ

- $\gamma = 4$ (mostly used)
- $\gamma = 4 \Rightarrow \text{ROUGE-2}$
- Bigram counts in a window of size n :

$$\text{Count}(k, n) = C \binom{n}{k} - \sum_0^{k-\gamma} (k - \gamma) ; \gamma \geq 1$$

ROUGE-n

- Exhaustive character
- Limited by content representation
- “blue convertible car” vs. “blue automobile”
- “The airplane A380” vs. “The Airbus 380 aeroplane”
- Possible to cheat for high ROUGE scores.
 - Sjöbergh 2007 study
 - Markov automata, bigram frequencies only.

ROUGE-n

- Exhaustive cl
- Limited by cc
- “blue convert
- “The airplane
- Possible to cl
 - Sjöbergh 2
 - Markov au

of the hurricane andrew had been injured and the storm caused by the gulf of mexico and louisiana and at least 150 000 people died on the us insurers expect to the florida and there are likely to be concentrated among other insurers have been badly damaged a result of damage caused in the state and to dollars 20bn of new orleans then to pay out on monday and new iberia associated tornadoes devastated inhabitants of miami

⋮

industry is the costliest disaster in florida as a quarter of hurricane hugo which hit the industry analysts cautioned that the bahamas on tuesday night or shut down

Evaluation aspects and metrics

Aspect	Metric	Explanation
Accuracy	ROUGE	Measures text overlap
	BLEU	Measures text overlap and computes precision
	METEOR	Measures text overlap including synonyms, etc.
	BERTScore	Measures cosine similarity between embeddings of summary and article
Faithfulness	SummaC	Computes alignment between individual sentences of summary and article
	QAFactEval	Verifies consistency of summary and article based on question answering
Compression	Compresion ratio	Measures ratio of number of tokens (words) in summary and article
Extractiveness	Coverage	Measures the extent to which summary text is from article
	Density	Quantifies how well the word sequence of a summary can be described as a series of extractions
Efficiency	Computation time	-
	Computation energy	-

Live Summarization Systems

<https://www.tools4noobs.com/summarize/>

<http://lexrank.herokuapp.com/index.html>

<http://proceedings.mlr.press/v119/zhang20ae.html>

Evaluation:

S. Shahbaz, D. Schwabe, and M. Potthast. Summary Workbench: Unifying Application and Evaluation of Text Summarization Models, (EMNLP 2022, arXiv)

Remarks on LLM/GPT Summaries

- Ubiquitous
- Build on classical methods + Transformer powers
- Increase science literacy
- Some authors find them to be on par with human summaries
 - (may be true for generic domains)

Remarks on LLM/GPT Summaries

- Ubiquitous
- Build on classical methods
- Increase science literacy
- Some authors find them
• (may be true for general public)

 SEMANTIC SCHOLAR

scientific summarization LLM

Reader's digest version of scientific writing: comparative evaluation of summarization capacity between large language models and medical students in analyzing scientific writing in sleep medicine

Jacob MatalonAugust SpurzemSana AhsanElizabeth WhiteRonik KothariMadhu Varma

Medicine, Computer Science · Frontiers Artif. Intell. · 24 December 2024

TLDR Comparing **summarization** capacity between LLM generated summaries of sleep medicine re by Medical Student (humans) and to evaluate if the re is retained comparably indicates that more re automatically generated short summary of the one strengths and weaknesses of AI models in co urate summaries.[Expand](#)

If you have feedback on this experience, [contact us](#).

TLDR LLM based approach to Scientific Text Simplification

Syed Muhammad AliHammad SajidOwais AijazOwais WaheedFaisal AlviAbdul Samad

Computer Science · [Conference and Labs of the Evaluation Forum](#) · 2024

TLDR Habib University's Team Sharingans' participation in the CLEF 2024 SimpleText track, which aims to simplify **scientific** texts for improved readability and comprehension for non-experts, is reported, proposing using fine-tuned large language models as a reliable source for **scientific** education.[Expand](#)

[PDF](#) [View Paper](#) [Save](#) [Cite](#)

Remarks on LLM/GPT Summaries

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The screenshot shows the ACM Digital Library interface. At the top, there are logos for ACM Digital Library and the Association for Computing Machinery. Below the logos is a navigation bar with links to Journals, Magazines, Proceedings, Books, SIGs, Conferences, Institutions, and People. A secondary navigation bar includes links to Conference, Proceedings, Upcoming Events, Authors, and Affiliations. The main content area displays the breadcrumb trail: Home > Conferences > SIGIR-AP > Proceedings > SIGIR-AP 2024 > A Reproducibility and Generalizability Study of Large Language Models for Query Generation. The article is labeled as a RESEARCH-ARTICLE with OPEN ACCESS. It features social media sharing icons for X, LinkedIn, Reddit, Facebook, and Email. The title is 'A Reproducibility and Generalizability Study of Large Language Models for Query Generation'. The authors listed are Moritz Staudinger, Wojciech Kusa, Florina Piroi, Aldo Lipani, and Allan Hanbury, with a link to 'Authors Info & Claims'. The publication details are: SIGIR-AP 2024: Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, Pages 186 - 196, with a DOI link. The publication date is 08 December 2024, and there is a 'Check for updates' button. Below the title, there are citation counts (6 citations, 853 views) and buttons for PDF and eReader. The abstract section is visible, showing the title 'Abstract' and the start of the text: 'Systematic literature reviews (SLRs) are a cornerstone of academic research, yet they are often labour-intensive and time-consuming due to the detailed literature curation process. The advent of generative AI and large language models (LLMs) promises to revolutionize this process by assisting researchers in several tedious tasks, one of them being the generation of effective Boolean queries that will select the publications to consider including in a review. This paper presents an extensive study of Boolean query generation using LLMs for systematic reviews, reproducing and extending the work of Wang et al. and Alaniz et al. Our'.

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A Reproducibility and Generalizability Study of Large Language Models for Query Generation

Authors: Moritz Staudinger, Wojciech Kusa, Florina Piroi, Aldo Lipani, Allan Hanbury | [Authors Info & Claims](#)

SIGIR-AP 2024: Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region
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Published: 08 December 2024 [Publication History](#)

6 853

Abstract

Systematic literature reviews (SLRs) are a cornerstone of academic research, yet they are often labour-intensive and time-consuming due to the detailed literature curation process. The advent of generative AI and large language models (LLMs) promises to revolutionize this process by assisting researchers in several tedious tasks, one of them being the generation of effective Boolean queries that will select the publications to consider including in a review. This paper presents an extensive study of Boolean query generation using LLMs for systematic reviews, reproducing and extending the work of Wang et al. and Alaniz et al. Our

Remarks on LLM/GPT Summaries

Issues:

- Omit details -> showing wrong research conclusions
- Overgeneralization
- Factual errors (LLM-generated content flows into new GPT/* models)

Remarks on LLM/GPT Summaries

Issues:

- Omit details -> showing wrong research conclusions
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Tested 10 prominent LLMs

- Generic generalizations
- Present tense generalization
- Action guided generalizations (impact on policy recommendations)

Remarks on LLM/GPT Summaries

Tested 10 prominent LLMs

- Non-generic to Generic
- Present tense generalization
- Action guided generalizations (impact on policy recommendations)

Original (26): 'Among patients undergoing hip or knee arthroplasty for osteoarthritis, **aspirin compared with enoxaparin resulted in a significantly higher rate of symptomatic VTE within 90 days**, defined as below- or above-knee DVT or pulmonary embolism'

LLaMA 3.3 70B (API): 'Overall, the study suggests that **enoxaparin is more effective than aspirin in preventing symptomatic VTE after THA or TKA**'

Remarks on LLM/GPT Summaries

Tested 10 prominent LLMs

- Non-generic to Generic
- Past tense to Present tense generalization
- Action guided generalizations (impact on policy recommendations)

Original (77): 'The consumption of caffeinated coffee **did not result in significantly more** daily premature atrial contractions than the avoidance of caffeine'

ChatGPT-3.5 (UI): 'The results suggest that the consumption of caffeinated coffee **does not significantly increase** the occurrence of premature atrial contractions'

Remarks on LLM/GPT Summaries

Tested 10 prominent LLMs

- Non-generic to Generic
- Present tense generalization
- Action guided generalizations (impact on policy recommendations)

Original (41): 'Our results are relevant to clinical practice, **supporting the use of the Mediterranean diet** in secondary prevention'

Claude 3.5 Sonnet (UI): 'The findings suggest that the **Mediterranean diet should be considered** as a preferred dietary approach for patients with established cardiovascular disease, particularly for men'

Content

- Introduction
- Processing & baseline algorithms
- Evaluation



Not covered: Abstracting text, Domain-specific summarization, query-guided summarization...

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